

Simulation Methods

Lecture Notes

Overview

- In many structural models, the likelihood or moment conditions **cannot be computed in closed form**
 - High-dimensional integrals over unobserved heterogeneity, latent states, or expectations over future shocks
 - Example: computing the likelihood of a consumption path in the savings model requires integrating over sequences of income shocks
- **Simulation** provides a way forward when analytical evaluation is intractable
- Motivating example: mean assets $\mathbb{E}[A_t]$ depends on the full history of income shocks $\varepsilon_1, \dots, \varepsilon_{t-1}$
 - With K_ε grid points, must sum over K_ε^{t-1} sequences—intractable for large t

Three Topics

1. **Simulation-based estimators:** MSM and indirect inference—replace intractable analytical objects with simulation approximations
 2. **The bootstrap:** resampling-based inference when analytical standard errors are difficult
 3. **Applications** to our prototype models
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Integration by Simulation

- Need to evaluate:

$$f(y|\mathbf{x}, \theta) = \int h(y|\mathbf{x}, \theta, \mathbf{u})g(\mathbf{u})d\mathbf{u}$$

where $\mathbf{u}$ is a vector of unobservables with density g

- Quadrature methods face the **curse of dimensionality** when $\mathbf{u}$ is high-dimensional
- **Monte Carlo integration:** draw $\mathbf{u}^1, \dots, \mathbf{u}^R$ from g and compute:

$$\hat{f}(y|\mathbf{x}, \theta) = \frac{1}{R} \sum_{r=1}^R h(y|\mathbf{x}, \theta, \mathbf{u}^r)$$

- Properties:
 - **Unbiased:** $\mathbb{E}_{\mathbf{u}}[\hat{f}] = f$
 - **Consistent:** $\hat{f} \rightarrow_p f$ as $R \rightarrow \infty$ by the LLN

- Convergence rate: $O(1/\sqrt{R})$ regardless of the dimension of $\mathbf{u}$ —this is the key advantage over quadrature

Importance Sampling

- Sometimes want to draw from an alternative density $w(\mathbf{u})$ instead of $g(\mathbf{u})$
 - g may be hard to sample from
 - Draws from w may reduce variance
- Importance sampling estimator:

$$\hat{f}(y|\mathbf{x}, \theta) = \frac{1}{R} \sum_{r=1}^R h(y|\mathbf{x}, \theta, \mathbf{u}^r) \frac{g(\mathbf{u}^r)}{w(\mathbf{u}^r)}, \quad \mathbf{u}^r \sim w$$

- The ratio g/w is the **importance weight**
- Still unbiased and consistent

Variance Reduction

- Variance depends on the variance of $h \cdot g/w$ under draws from w
- If w concentrates draws where the integrand $h \cdot g$ is large, fewer draws are “wasted” on low-contribution regions
- Ideal importance density: proportional to $|h| \cdot g$ —puts weight where it matters most
- In practice: find a convenient w that roughly approximates this shape

Example: Tail Probability

- Estimating $P(Z > 3)$ where $Z \sim \mathcal{N}(0, 1)$ (true value ≈ 0.00135)
- **Naïve MC**: most draws contribute zero (event is rare)—highly variable, discrete
- **Importance sampling** with $w = \mathcal{N}(3, 1)$: concentrates draws in the tail
- Result: both unbiased, but importance sampling achieves dramatic variance reduction

Differentiability

- Importance sampling also helps make simulated integrals **differentiable** in θ
- When integrand involves indicators $\mathbf{1}\{\cdot\}$ (discrete choice models), crude MC gives a step function in θ
- Through change of variables, can move θ -dependence from indicator into smooth importance weights
- The **GHK simulator** for multivariate probit is a prominent example

Maximum Simulated Likelihood

- Replace the likelihood with its simulated analogue:

$$\hat{\theta}_{MSL} = \arg \max_{\theta} \sum_{n=1}^N \log \hat{f}(y_n|\mathbf{x}_n, \theta)$$

Key Subtlety: Bias from the Logarithm

- Because log is concave, Jensen's inequality gives:

$$\mathbb{E}[\log \hat{f}] \leq \log \mathbb{E}[\hat{f}] = \log f$$

- The simulated log-likelihood is **biased downward**
- MSL is **inconsistent for fixed R**
- Consistency requires both $N, R \rightarrow \infty$ with $\sqrt{N}/R \rightarrow 0$
- **Contrast with MSM:** MSM is consistent for any fixed R

Asymptotic Distribution

- Under regularity conditions and $\sqrt{N}/R \rightarrow 0$:

$$\sqrt{N}(\hat{\theta}_{MSL} - \theta_0) \rightarrow_d \mathcal{N}(\mathbf{0}, \mathcal{I}(\theta_0)^{-1})$$

- Same asymptotic distribution as infeasible MLE
- Rule of thumb: use enough draws that results are insensitive to further increases

Differentiability of the Simulated Likelihood

- **Problem:** when the model involves discrete outcomes, crude MC gives a step function in θ
 - Example: probit with $Y = \mathbf{1}\{X\beta + \varepsilon > 0\}$
 - True probability $\Phi(X\beta)$ is smooth, but $\hat{P} = \frac{1}{R} \sum_r \mathbf{1}\{X\beta + \varepsilon^r > 0\}$ is a step function
 - Derivative is zero almost everywhere—gradient-based optimization fails
- **Solutions:**
 1. **Kernel smoothing:** replace $\mathbf{1}\{\cdot > 0\}$ with $\Phi(\cdot/h)$ for small bandwidth h
 2. **Importance sampling / change of variables:** move θ into smooth weights (GHK simulator)
 3. **Derivative-free optimization:** Nelder-Mead (simple but slow/unreliable in high dimensions)

Method of Simulated Moments

- The simulation analogue of GMM
- Moment conditions:

$$\mathbb{E}[g(\mathbf{w}, \theta_0)] = \mathbb{E} \left[\int h(\mathbf{w}, \theta_0, \mathbf{u}) f(\mathbf{u}|\theta_0) d\mathbf{u} \right] = \mathbf{0}$$

- Replace with simulated version:

$$\hat{g}(\mathbf{w}_n, \theta) = \frac{1}{R} \sum_{r=1}^R \tilde{g}(\mathbf{w}_n, \mathbf{u}^r, \theta)$$

where $\mathbb{E}_{\mathbf{u}}[\tilde{g}] = g$

- MSM estimator:

$$\hat{\theta}_{MSM} = \arg \min_{\theta} \hat{\mathbf{g}}_N(\theta)' \mathbf{W} \hat{\mathbf{g}}_N(\theta), \quad \hat{\mathbf{g}}_N(\theta) = \frac{1}{N} \sum_{n=1}^N \hat{g}(\mathbf{w}_n, \theta)$$

Common Special Case

- In many structural applications, match **aggregate** statistics:

$$\hat{\mathbf{g}}_N(\theta) = \hat{\mathbf{m}} - \tilde{\mathbf{m}}(\theta)$$

- $\hat{\mathbf{m}}$: vector of sample statistics (means, variances, etc.)
- $\tilde{\mathbf{m}}(\theta)$: corresponding vector from simulated data
- This is the structure used for the savings model

Asymptotic Distribution

- Under standard GMM regularity conditions with R fixed:

$$\sqrt{N}(\hat{\theta}_{MSM} - \theta_0) \rightarrow_d \mathcal{N}(\mathbf{0}, \Sigma_{MSM})$$

where:

$$\Sigma_{MSM} = (\nabla_{\theta} g_0 \mathbf{W} \nabla_{\theta} g_0')^{-1} \nabla_{\theta} g_0 \mathbf{W} \mathbb{E}[\hat{g}_0 \hat{g}_0'] \mathbf{W} \nabla_{\theta} g_0' (\nabla_{\theta} g_0 \mathbf{W} \nabla_{\theta} g_0')^{-1}$$

Key Properties

1. **Consistency does not require** $R \rightarrow \infty$: because $\hat{g}$ is unbiased for g , sample moments converge at θ_0 by LLN. Contrast with MSL.
2. **Simulation adds variance**: $\mathbb{V}[\hat{g}] \geq \mathbb{V}[g]$, so MSM is less efficient than infeasible GMM. For frequency simulators, variance inflates by factor $(1 + 1/R)$.
3. **If** $\sqrt{N}/R \rightarrow 0$: efficiency loss disappears and MSM achieves same asymptotic variance as GMM.

Savings Model Application

- Estimate preference parameters (β, σ) with $\psi = 0$ (no bequest motive)
- Simulate life-cycle profiles: solve DP, draw income shocks, track assets/consumption
- Target moments: mean assets and variance of log consumption at ages 30, 40, 50, 60
- Use Nelder-Mead (derivative-free) because discrete asset grid makes simulated moments non-differentiable

Indirect Inference

- Fit an **auxiliary model** to both real data and simulated data, then find structural parameters that make the two sets of auxiliary estimates match

Setup

- $\hat{\beta}_N$: auxiliary parameters estimated from data (e.g. OLS coefficients, reduced-form probit)
 - Asymptotically normal: $\sqrt{N}(\hat{\beta}_N - \beta_0) \rightarrow_d \mathcal{N}(\mathbf{0}, \Omega)$
- For each candidate θ : simulate data, compute same statistic $\hat{\beta}^R(\theta)$
- $\beta(\theta) = \text{plim } \hat{\beta}^R(\theta)$: pseudo-true auxiliary parameter implied by θ
- **Indirect inference estimator**:

$$\hat{\theta}_{II} = \arg \min_{\theta} (\hat{\beta}_N - \hat{\beta}^R(\theta))' \mathbf{W} (\hat{\beta}_N - \hat{\beta}^R(\theta))$$

Asymptotic Distribution

- Under identification (θ_0 uniquely solves $\beta(\theta) = \beta_0$) and $\sqrt{N}/R \rightarrow 0$:

$$\sqrt{N}(\hat{\theta}_{II} - \theta_0) \rightarrow_d \mathcal{N}(\mathbf{0}, (\nabla_{\theta}\beta_0 \mathbf{W} \nabla_{\theta}\beta_0')^{-1} \nabla_{\theta}\beta_0 \mathbf{W} \Omega \mathbf{W} \nabla_{\theta}\beta_0' (\nabla_{\theta}\beta_0 \mathbf{W} \nabla_{\theta}\beta_0')^{-1})$$

- Same sandwich structure as minimum distance—indirect inference **is** a simulated minimum distance estimator

Relationship to MSM

- Indirect inference **neests MSM** as a special case
 - If auxiliary model consists of sample moments, indirect inference reduces to MSM
- The appeal: allows **richer auxiliary models** (regressions, multinomial choice) that capture complex data features

STOP FOR DISCUSSION

Choosing auxiliary statistics:

- A good auxiliary model should be **sensitive** to the structural parameters of interest
- Should capture the key features the structural model is designed to explain
- Should be computationally cheap (must be re-estimated on each simulated dataset)
- Think of auxiliary model as a **diagnostic**: if the structural model cannot match auxiliary estimates, it suggests the model is missing something important

The Bootstrap

- A fundamentally different use of simulation: not for evaluating objective functions, but for **inference**
- Uses resampling to approximate the sampling distribution of an estimator

Motivation

- When do we need the bootstrap?
 1. Variance formula is difficult to derive (e.g. multi-step estimators)
 2. Derivatives of the criterion function are hard or costly to compute
 3. Estimation criterion is non-smooth in finite samples
 4. Sample size too small for CLT approximation
- The bootstrap addresses all of these by directly approximating the finite-sample distribution

Framework

- Let $T_N = T_N(X_1, \dots, X_N)$ be a statistic (e.g. $\hat{\theta}$ or $\sqrt{N}(\hat{\theta} - \theta_0)$)
- Finite-sample distribution $G_N(\cdot, F_0)$ is typically unknown
- **Bootstrap idea**: replace F_0 with estimate $\hat{F}_N$ and use $G_N(\cdot, \hat{F}_N)$

Two Main Approaches

1. **Nonparametric bootstrap**: replace F_0 with the empirical distribution
 - Bootstrap samples: N observations drawn **with replacement** from original data

- More general, fewer assumptions
2. **Parametric bootstrap**: replace F_0 with $F(\cdot, \hat{\theta}_N)$
 - Bootstrap samples: **simulate** from the estimated model
 - More efficient when model is correctly specified
 - Natural for structural models (already have simulation machinery)

Procedure

Given estimator $\hat{\theta}_N$:

1. Draw bootstrap sample $\mathbf{X}^b$ (resample or simulate)
2. Compute bootstrap estimate $\hat{\theta}_N^b = T_N(\mathbf{X}^b)$
3. Repeat for $b = 1, \dots, B$
4. From the bootstrap distribution $\{\hat{\theta}_N^1, \dots, \hat{\theta}_N^B\}$ compute:
 - **Standard errors**: $\text{se}^*(\hat{\theta}) = \text{sd}(\hat{\theta}^1, \dots, \hat{\theta}^B)$
 - **Percentile CIs**: $[q_{\alpha/2}, q_{1-\alpha/2}]$
 - **Bootstrap- t CIs**: based on distribution of $t^b = (\hat{\theta}^b - \hat{\theta})/\text{se}(\hat{\theta}^b)$

When Does the Bootstrap Work?

- Consistent when T_N is asymptotically normal
- Can do **better** than asymptotic normal approximation when statistic is **asymptotically pivotal** (distribution doesn't depend on unknowns)
 - The t -statistic is a canonical example
 - Bootstrap provides **higher-order refinement** over normal approximation

Replicating the Dependence Structure

- Resampling must correctly replicate the data's dependence structure:
 - **iid data**: sample with replacement
 - **Panel data**: resample **entire individuals** (not individual observations) to preserve within-person correlation
 - **Time series**: block bootstrap methods

Income Process Application

- Apply bootstrap to the income process parameters estimated by minimum distance
 - Resample **persons** (not person-year observations) from the PSID panel
 - For each bootstrap sample: re-estimate $(\rho, \sigma_\alpha^2, \sigma_\eta^2)$
 - Automatically accounts for within-person dependence, heteroskedasticity, and complexity of the estimator
 - Compare bootstrap SEs to analytical SEs from the previous chapter
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Practical Considerations

Simulation Noise and Smoothing

- **Always fix the random seed** when evaluating the simulated objective across different θ values
- With fixed draws: simulated objective is a smooth function of θ (if model is smooth)
- With re-drawn draws: objective becomes noisy, optimization may fail or converge incorrectly

Simulation Noise and Standard Errors

- Must think carefully about seeds when using bootstrap for inference
- If R is comparable to N : simulation error is a non-negligible variance component
- **Change the initial seed** in each bootstrap trial
- Even if $R \gg N$: good practice and costless to incorporate simulation error

How Many Draws?

- **MSM**: consistency holds for any fixed R ; efficiency improves with R
 - Start moderate, increase and check stability
- **MSL**: bias from log of simulated probabilities is $O(1/R)$
 - R should be large enough that bias is negligible relative to standard error

Choosing Moments for MSM

1. **Identification**: moments should be informative about parameters
2. **Parsimony**: more moments improve efficiency in principle but increase cost and can lead to poorly conditioned weighting matrices
3. **Model fit**: if model cannot match a moment, including it distorts other estimates

Diagonal Weighting Matrices

- Optimal $\mathbf{W}^* = \hat{S}^{-1}$ can be poorly estimated with many moments
- Practical alternative: **diagonal** weighting matrix (inverse variance of each moment, ignoring cross-moment covariances)
- Sacrifices some efficiency but much more stable in practice

STOP FOR DISCUSSION

- Why is it critical to fix the simulation seed *during* optimization but change it *across* bootstrap replications?
- What would happen if you failed to fix the seed inside the optimization routine?

Non-Differentiability of Simulated Moments

- In the savings model, the policy function maps to a **discrete** asset grid
- Simulated consumption and asset paths are step functions of θ
- As θ changes continuously, optimal grid point occasionally jumps $\Rightarrow$ discontinuities in simulated moments
- Same non-differentiability problem as in MSL with discrete outcomes

- **Gradient-based optimizers fail:** numerical gradient is zero almost everywhere
- **Solutions:**
 - Derivative-free methods (Nelder-Mead)
 - Finer asset grid (reduces jump size)
 - Interpolating policy function between grid points
 - Kernel-smoothed simulators